A.D.A.P.T.: A Weather-Informed, Verifiable Pipeline from Annotated Flood Maps to Executable UAV Relief Missions

Abstract

Flood relief requires converting mapped hazards and delivery points into valid UAV missions. We evaluate A.D.A.P.T., which processes annotated flood rasters and weather, selects dry relief drops along hazard perimeters, routes a battery-constrained multi-base drone fleet, and exports executable WPL 110 missions. Its segmentation uses operator-sampled HSV thresholds; its forecast is a terrain-free advection-dilation spread proxy. The multi-base engine samples candidate launch sites, computes shortest-path distance fields, selects up to K bases, assigns drops by reachability, partitions routes into battery-limited sorties with reload stops, and checks for inter-UAV route crossings. On demonstration benchmarks, all fleet missions pass an independent structural validator. On 88 held-out Sen1Floods11 chips, S1 Otsu, a small S1 U-Net and S2 NDWI achieve micro IoU 0.225, 0.606 and 0.772. A GeoTIFF-referenced diagnostic mission avoids a 24.5 m scalar-coordinate error. Across two observed flood transitions, the spread proxy ties persistence. ArduCopter SITL and MAVProxy accept and execute the exported multi-waypoint missions. The multi-base framework scales relief coverage with base cap K while enforcing strict dry-drop and endurance constraints.

Keywords: UAV relief, flood mapping, mission planning, multi-UAV allocation, geospatial verification, WPL 110.

1 Introduction

Floods interrupt roads when isolated communities need supplies. UAVs can map affected areas and carry small urgent payloads [1,2]. Flood extents are available in satellite and aerial products [3, 4], yet a relief mission still requires an operator to identify drops, order visits, avoid hazards, convert image locations to coordinates, and encode autopilot commands. The reviewed segmentation, routing and inundation studies usually stop at one of these stages [5–7]. Mission generation and its geographic correctness deserve evaluation as part of the chain.

1.1 Motivation and gap

A.D.A.P.T. accepts an annotated flood-map raster, weather and a geographic reference. Its multi-base planning engine converts detected flood regions into executable QGroundControl WPL 110 mission files for an autonomous fleet of up to K UAVs. Existing perception methods often stop at a segmentation mask, while routing studies usually assume pre-located demand points. The capability matrix in Table 1 shows the narrower gap addressed here: an end-to-end image-to-fleet-mission pipeline with energy-aware sortie scheduling, dry-zone drop placement, and independently checked coordinate and autopilot mission layers.

1.2 Contributions

The paper makes three checkable contributions. First, it specifies and evaluates the end-to-end pipeline from operator-sampled flood rasters through dynamic weather-driven hazard modeling to multi-base fleet missions (Sections 3–4). Second, it formulates and evaluates the multi-base fleet planner under battery endurance, payload capacity, dry-ground clearance and geometric route-crossing constraints, evaluating scaling behavior from $K = 1$ to multiple bases on benchmark maps (Section 4). Third, it separates software correctness from geographic and operational validity using an independent WGS84 geodesic oracle, a specification-based WPL validator, adversarial stress tests and hardware-in-the-loop SITL simulations (Section 6).

The remainder reviews related work, formalises the problem and architecture, details the multi-base fleet methodology and verification, reports experimental results, and closes with limitations and future work.

2 Related Work

2.1 UAVs in Disaster Response

Surveys of UAV use in disasters converge on three roles: rapid mapping and damage assessment, communication support, and delivery of small, time-critical payloads [1,2,8]. Erdelj *et al.* [1] frame these roles across the prediction, assessment and response phases, and humanitarian-logistics reviews confirm that delivery to areas cut off by floods is a recurring motivation [9]. Most of this literature is either conceptual or concerned with networking and coordination; for example, multi-UAV search-and-rescue frameworks address task allocation among vehicles [10], and recent flood-management work simulates UAV-cluster deployment and resource scheduling with edge computing [11]. Gioia *et al.* [12] review adaptive mission planning and hazard monitoring for wildfires, chemical releases and floods, and identify the integration of hazard-evolution models with mission planning as fragmented across domains. These works motivate the problem but do not provide a path from a flood map to a mission file.

2.2 Flood Mapping and Segmentation

Classical water mapping relies on spectral indices such as the NDWI [13] and, for all-weather operation, on synthetic aperture radar with automated thresholding chains [3]. Learning-based methods require labelled data; Sen1Floods11 provides a georeferenced benchmark for Sentinel-1 [4], and FloodNet provides pixel-labelled post-flood UAV imagery [14]. Convolutional networks have been applied to UAV flood-extent mapping [5]. Training-free colour-based methods remain attractive where labels are unavailable: Simantiris and Panagiotakis [15] combine LAB-colour masks, a vegetation index and edges for unsupervised flood segmentation of UAV photographs, and Ibrahim *et al.* [16] compare RGB/HSI clustering and region growing. All of these methods terminate at a flood mask; none produces delivery points or a flight plan. A.D.A.P.T. addresses a narrower perception problem than these works: it extracts flood pixels from *already annotated* map products by operator-sampled HSV thresholding [17], followed by mathematical morphology [18] and border-following contour extraction [19].

2.3 Flood Modelling and Prediction

Physically based and simplified inundation models range from raster storage-cell schemes such as LISFLOOD-FP [20] to cellular-automata (CA) models that replace the shallow-water equations by local transition rules on a digital elevation model [21–23]. CA models trade some accuracy for large speed-ups, but still require terrain data, rainfall or inflow forcing and calibration; Jamali *et al.* [23] also report weaker performance where flow momentum matters. Teng *et al.* [6] review this spectrum and its uncertainties. No reviewed model is coupled to UAV mission planning. The spread rule in A.D.A.P.T. is not a hydrological model: it is a terrain-free morphological proxy that

dilates and shifts the observed mask according to precipitation and wind, and it is used only to inflate obstacles.

2.4 Path Planning

Grid search with A* [24] is optimal for static maps. Incremental methods, D* [25] and its simpler successor D* Lite [26, 27], reuse search effort when edge costs change, which is valuable when a vehicle discovers obstacles during execution. Reviews of UAV path planning [28] catalogue classical, sampling-based and learning-based planners; they assume that the obstacle representation is given. A.D.A.P.T. uses D* Lite on a downsampled occupancy grid derived from current and predicted flood masks; as shown in Section 8, its present offline use does not exercise the incremental property.

2.5 Route Optimisation and Relief Logistics

Visiting a set of delivery points is a travelling-salesman-type problem. Exact dynamic programming is exponential [29], and the nearest-neighbour construction heuristic can be a logarithmic factor worse than optimal in the worst case [30]. Operations-research work on UAV delivery formulates truck–drone tandems [31] and drone fleets for last-mile relief [32]; Otto *et al.* [7] survey this area. Most closely related, Dong *et al.* [33] optimise multi-vehicle, multi-UAV flood-rescue delivery with flood-depth-dependent road speeds using mixed-integer programming and adaptive large-neighbourhood search. These models are more rigorous than A.D.A.P.T.'s ordering heuristic, but take demand locations and network states as given inputs rather than deriving them from imagery, and output schedules rather than autopilot missions.

2.6 Landing and Drop-Site Selection

Image-based selection of safe landing zones has been studied using multichannel imagery fused with map data [34] and texture-based flat-surface detection [35]. These methods score candidate sites for landing safety. A.D.A.P.T. has no comparable safety model: its "safe" drop point is a convex-hull vertex on the boundary of a flood region, chosen for accessibility from the route, without terrain, population or obstacle information.

2.7 Mission Generation and Geospatial Conversion

MAVLink is the de facto protocol between ground stations and open autopilots [36, 37]; missions are commonly exchanged as QGroundControl WPL plain-text files [38] whose command semantics are defined by the common message set [39] and refined per autopilot [40,41]. Converting image coordinates to geodetic coordinates requires a projection model [42]; web-map imagery uses Web Mercator, whose scale varies with latitude [43], and accurate geodesics on the ellipsoid are available from Vincenty's and Karney's algorithms [44, 45]. We found no reviewed work that generates WPL missions from flood imagery and verifies the geodetic and mission layers.

2.8 Synthesis and Gap

Table 1 summarises the capabilities reported by representative work. Perception, hazard modelling, path planning, route optimisation and site selection are each well developed, but in the reviewed literature they are rarely combined into a chain that starts from flood imagery and ends in an executable autopilot mission, and the correctness of the image-to-geodetic and mission-encoding steps is not reported. A.D.A.P.T. addresses this intersection with deliberately simple components. Its contribution is integration and verification; each of its components is weaker than the specialised state of the art in its own area.

3 Problem Formulation and Architecture

3.1 Notation

Let I be the input raster of $W_0 \times H_0$ pixels. If W_0 exceeds a display width D (750 px by default), the image is resized to I' of size $W' \times H'$ with $W' = D$, $H' = \mathrm{round}(H_0 s)$ and $s = D/W_0$; otherwise $I' = I$ and $s = 1$. All perception and planning operate on the pixel domain $\Omega = \{0, \ldots, W'-1\} \times \{0, \ldots, H'-1\}$ of I', with OpenCV conventions: x is the column (increasing to the east of a north-up map) and y the row (increasing to the south), and integer coordinates denote pixel centres.

The principal symbols and outputs are in Table 2.

3.2 Problem Statement

Given a flood map raster I, local weather $w = (P, V, \theta)$, geographic anchor $(\varphi_0, \lambda_0, \rho)$ and a maximum base cap K, synthesize executable mission files $\mathcal{W}_1, \ldots, \mathcal{W}_m$ $(m \leq K)$ for an autonomous drone fleet. The fleet must:

1. operate from flood-clear launch bases $\{b_1, \ldots, b_m\}$ established on dry ground with a safety clearance from water;
2. deliver relief payloads to dry drop points $\{d_i\}$ placed along the unflooded perimeters of isolated flood regions;
3. strictly avoid dynamic flood obstacles $O = M \cup \hat{M}$ formed by current inundation and near-term weather-driven spread;
4. satisfy vehicle payload limits and battery endurance limits with a mandatory reserve margin r;
5. ensure that no two vehicles have crossing flight trajectories.

Setting $K = 1$ specifies the single-base case, where a single UAV serves the reachable cluster of drops within its battery limit.

3.3 Architecture and Interfaces

Fig. 1 illustrates the implemented end-to-end data flow. The entry point validates its arguments and geographic parameters before execution, then executes the pipeline in four decoupled stages:

1. *Perception and hazard modeling:* Color sampling and morphological filtering extract current flood mask M and region polygons R_i. Weather conditions drive an advection-dilation model to yield predicted spread $\hat{M}$, giving the composite obstacle field $O = M \cup \hat{M}$.
2. *Dry drop placement:* Drops are positioned on dry perimeter bands beside each hazard region according to a tiered area-based rule.
3. *Fleet planning and routing:* Candidate launch sites are generated outside O. Shortest-path distance fields δ are computed on a conservative grid. A greedy scoring routine selects up to K bases and assigns each drop to its nearest feasible base. Sorties are planned with battery replenishment and payload reloads, followed by an inter-mission crossing check.
4. *Geodetic mission synthesis and verification:* Waypoints are transformed from pixel space to WGS-84 coordinates and exported as standard QGroundControl WPL 110 files. All files undergo specification-based syntax and geodesic validation.

Each run is isolated in a timestamped session directory, preventing stale file reuse. If unresolvable route crossings or unreachable drops occur, the planner logs explicit safety status diagnostics.

Table 1: Capabilities reported by representative related work and by A.D.A.P.T.. ✓: addressed; (✓): partially or under strong simplification; –: not addressed in the cited work as reported. The table reflects the stated scope of each paper and is not exhaustive.

Work (group)	Flood perception	Hazard evolution	Weather input	Obstacle-aware path planning	Delivery-point selection	Visit ordering	Executable autopilot mission	Geo/mission verification
Flood segmentation [5, 14, 15]	✓	–	–	–	–	–	–	–
SAR flood mapping [3, 4]	✓	–	–	–	–	–	–	–
Inundation models [20–23]	–	✓	(✓)	–	–	–	–	–
Incremental grid planning [25, 27]	–	–	–	✓	–	–	–	–
Relief / drone routing [31, 32]	–	–	–	–	–	✓	–	–
Flood-rescue collaborative delivery [33]	–	–	–	(✓)	–	✓	–	–
Safe landing-zone detection [34, 35]	(✓)	–	–	–	(✓)	–	–	–
Multi-UAV SAR / flood deployment [10, 11]	–	–	–	(✓)	–	✓	–	–
A.D.A.P.T. (this work)	(✓)[a]	(✓)[b]	✓[c]	✓	(✓)[d]	(✓)[e]	✓	✓[f]

[a] Annotated map products only, operator-sampled colour thresholds. [b] Terrain-free morphological proxy, uncalibrated. [c] Current precipitation and wind only. [d] Boundary geometry only, no safety model. [e] Nearest-neighbour heuristic; the optional multi-base mode assigns drops to up to K bases by a nearest-feasible rule under a battery model. [f] Software verification and one ArduCopter SITL check; no ground-station GUI or physical flight validation.

Table 2: Symbols and parameters used in the multi-base fleet planning architecture.

Symbol	Meaning
I, I'	Input raster and resized display image; $s = D/W_0$ is their scale.
$M, \hat{M}, O$	Detected flood, predicted flood, and planning obstacles $M \cup \hat{M}$.
P, V, θ	Precipitation, wind speed, and meteorological wind direction.
$R_i, d_{i,j}, b_k$	Flood region, assigned dry drop point, and selected launch base.
$G, \delta, \mathcal{S}_k$	Conservative routing grid, shortest-path distance field, and sorties.
K, E, r	Base cap, loaded endurance, and reserved battery fraction.
$\varphi_0, \lambda_0, \rho$	Centre latitude/longitude and metres per original image pixel.
$\mathcal{W}_k$	QGC WPL 110 mission item sequence for vehicle k.

4 Flood Perception and Environmental Hazard Modeling

This section details the perception and environmental modeling stages of A.D.A.P.T. (Fig. 2). All operations operate on the display domain $\Omega = \{0, \dots, W' - 1\} \times \{0, \dots, H' - 1\}$ using the repository configuration defaults (Table 5).

4.1 Input Acquisition and Display Scaling

The input is a flood-annotated map raster I in which flooded extents are demarcated. The image is loaded via OpenCV and scaled to display width $D = 750$ px using area interpolation: I' has width $W' = D$ and height $H' = \text{round}(H_0 s)$ with $s = D/W_0$ (Section 3). Scaling ensures efficient interactive sampling; because subsequent stages operate in display pixels, all pixel parameters have ground equivalents scaled by $k = \rho/s$, as handled in the geodetic conversion (Section 5.6).

4.2 Colour Sampling and Flood Segmentation

An operator clicks K_{click} flood pixels $c_k = (x_k, y_k)$ in I'. For each click, an 11×11 patch Π_k is extracted and mapped to HSV space [17] ($H \in [0, 179]$, $S, V \in [0, 255]$). Let $\mathcal{S} = \bigcup_k \text{HSV}(I'(\Pi_k))$. Denoting channel sample minima and maxima by C^- and C^+, thresholds are defined with margins $\delta_H = 20$ and $\delta_{SV} = 30$:

$$\ell_H = \max(H^- - \delta_H, 0), \quad u_H = \min(H^+ + \delta_H, 179), \tag{1}$$

$$\ell_C = \max(C^- - \delta_{SV}, 0), \quad u_C = \min(C^+ + \delta_{SV}, 255), \tag{2}$$

for $C \in \{S, V\}$. When $\ell_H \le 10$ or $u_H \ge 170$, the red-wrap rule sets $\mathcal{H} = [0, 10] \cup [170, 179]$; otherwise $\mathcal{H} = [\ell_H, u_H]$. The raw segmentation mask M_0 is filtered by morphological closing followed by opening [18] using a 5×5 square structuring element B:

$$M = (M_0 \bullet B) \circ B. \tag{3}$$

4.3 Flood-Region Extraction and Free Perimeter Bands

External contours of M are extracted via border following [19]. Contours with enclosed area below $a_{\min} = 200$ px^2 are discarded as noise. Each valid region R_i is represented by its polygon vertices, its filled binary area $\bar{R}_i$, and its centroid $c_i = (\lfloor m_{10}/m_{00} \rfloor, \lfloor m_{01}/m_{00} \rfloor)$. For relief operations, delivery points must be located on dry ground immediately adjacent to flood zones. Let Q_{2r+1} be a square structuring element of side $2r + 1$. The unflooded perimeter band of radius r around region $\bar{R}_i$ is defined by:

$$B_r = (\bar{R}_i \oplus Q_{2r+1}) \setminus O. \tag{4}$$

With $r_{\max} = \max(2, \min(20, \lceil 0.03 \max(W', H') \rceil))$, the search radius $r^* \le r_{\max}$ is the smallest integer for which $B_{r^*} \neq \emptyset$.

4.4 Weather Acquisition and Dynamic Spread Heuristic

Local weather at reference (φ_0, λ_0) is ingested from the Open-Meteo forecast API [46] or cached configuration: precipitation P (mm), wind speed V (km/h) and meteorological direction θ. Near-term hazard growth is modeled over planning horizon $T = 2$ h using an advection-dilation cellular heuristic. In each hourly iteration, the flood front dilates isotropically based on rainfall accumulation and shifts downwind along direction $\alpha = \theta + 180°$:

$$b = \max(0, \lfloor P/5 \rfloor), \quad v = \max(0, \lfloor V/10 \rfloor), \quad (\Delta x, \Delta y) = (v \sin\alpha, -v \cos\alpha \tag{5}$$

Applying $N = \max(1, \lfloor T \rfloor)$ iterations yields the predicted hazard mask $\hat{M}$. The composite obstacle field $O = M \cup \hat{M}$ forms the forbidden airspace for all flight paths.

4.5 Tiered Dry Relief Drop Placement

Unlike methods that place delivery waypoints directly on flood edges (which may be submerged during delivery), A.D.A.P.T. guarantees dry-ground drop points within the perimeter band B_{r^*}. To scale delivery capacity with hazard extent, regions receive multiple drop points based on enclosed area A:

$$n_i = \begin{cases} 1 & A < a_{\text{mod}}, \\ 2 & a_{\text{mod}} \le A < a_{\text{large}}, \\ 3 & a_{\text{large}} \le A < a_{\text{huge}}, \\ \min(k_{\text{drop}},\, 3 + \lfloor (A - a_{\text{huge}})/a_{\text{step}} \rfloor) & A \ge a_{\text{huge}}, \end{cases} \tag{6}$$

with thresholds $a_{\text{mod}} = 2000$ px^2, $a_{\text{large}} = 8000$ px^2, $a_{\text{huge}} = 25\,000$ px^2, step size $a_{\text{step}} = 3000$ px^2 and cap $k_{\text{drop}} = 4$. For each drop $j \in \{0, \dots, n_i - 1\}$, an anchor vertex at perimeter fraction j/n_i selects the nearest dry pixel in B_{r^*} that maintains a mutual separation of at least $s_{\text{drop}} = 50$ px from all earlier drops of that region. Drops unable to satisfy clearance are pruned, and single-point fallback ensures every validated region receives at least one relief delivery.

5 Multi-Base Fleet Planning and Mission Synthesis

This section specifies the autonomous multi-base planning, routing, and mission generation engine of A.D.A.P.T. (Fig. 3). Given the obstacle field O and dry drop points $\{d_i\}$, the planner selects up to K launch bases, assigns drops to bases by routed reachability, partitions routes into battery-feasible sorties, repairs route crossings, and exports validated WPL 110 autopilot missions.

5.1 Base Candidates and Routed Distance Fields

Candidate launch sites are sampled on a regular grid outside O with stride $\lfloor \max(W', H')/60 \rfloor$, retaining only sites with Euclidean clearance at least 5.0 px from any obstacle pixel. To ensure collision-free paths, distances are computed over a conservative routing grid where O is downsampled by factor $f = 3$; a cell is blocked if any pixel within its 3×3 block is occupied.

For every drop point d, an 8-connected shortest-path field is computed with step costs 1 (orthogonal) and $\sqrt{2}$ (diagonal). The routed distance $\delta(b, d)$ between a candidate base b and drop d is the field value multiplied by $3k$ metres ($k = \rho/s$); $\delta(b, d) = \infty$ if no obstacle-free path exists. Blocked coordinates are mapped to grid cells using a component-aware breadth-first search of radius at most 4 cells.

5.2 Battery Endurance and Payload Formulation

Battery expenditure is modeled as a fraction of loaded endurance $E = 7$ min at cruise speed $v = 5$ m/s. A flight segment of length ℓ carrying payload m against headwind $w = V/3.6$ m/s consumes:

$$\beta(\ell, m, w) = \frac{\ell}{60\, v\, E} \left(1 + c_m m + c_w w\right) \tag{7}$$

of full battery capacity, with mass coefficient $c_m = 0.08$ kg^{-1} and wind coefficient $c_w = 0.025$ s/m. Each delivery expends payload $m_d = 0.25$ kg from an onboard capacity of 8.0 kg. Every sortie must preserve a mandatory reserve margin of at least $r = 20\%$ battery upon return to base.

5.3 Base Selection and Reachability Assignment

A drop d is *reachable* from candidate site b if a round-trip single-drop sortie satisfies the usable battery budget:

$$\beta\big(\delta(b, d), m_d, w\big) + \beta\big(\delta(b, d), 0, w\big) \leq 1 - r. \tag{8}$$

Drops unreachable from all candidate sites are reported and excluded. If a single candidate reaches all coverable drops, that site is selected as the sole base ($K = 1$). Otherwise, a greedy procedure selects up to K bases (`MAX_BASES`, default 6) spaced at least 150 px apart. The procedure prioritizes bases that minimize a composite score comprising total estimated flight time and cumulative sortie battery fractions.

Each drop is assigned to the nearest reachable base by routed distance δ. Setting $K = 1$ operates the fleet engine as a single-UAV system with strict endurance and dry-ground constraints.

5.4 Closed-Loop Sortie Generation and Reloads

Each chosen base plans its deliveries independently:

1. *Sortie Partitioning:* Drops assigned to a base are visited in nearest-neighbour order. Drops are accumulated into a sortie as long as the cumulative round-trip flight (including return to base) satisfies remaining battery $\geq r$ and cumulative payload ≤ 8 kg. When a constraint is reached, the vehicle returns to base, executes a reload/recharge sequence, and initiates the next sortie.

2. *Time-Dependent Arrival Checks:* For each leg, arrival time t is estimated from cumulative distance. The dynamic obstacle boundary $F(t)$ is checked to ensure drop locations remain dry at estimated arrival time.

Table 3: Autopilot mission command sequence generated for each fleet UAV (QGC WPL 110 format; altitudes relative to launch base).

Item	MAVLink command (id)	Alt.	Role / Parameters
0	NAV_WAYPOINT (16)	0	Launch Base Home
1	NAV_TAKEOFF (22)	100	Climb to cruise altitude
For each route transit point:			
	NAV_WAYPOINT (16)	100	Cruise navigation
At each relief delivery stop:			
	NAV_LOITER_TIME (19)	100	p1 = 5 s loiter
	NAV_WAYPOINT (16)	10	Descend to drop height
	DO_SET_SERVO (183)	10	Ch 9, PWM = 2000 μs
	NAV_WAYPOINT (16)	100	Climb to cruise height
At each reload/recharge stop:			
	NAV_LAND (21)	0	Land at base
	NAV_TAKEOFF (22)	100	Resume next sortie
$m-1$	NAV_RETURN_TO_LAUNCH (20)	100	Return to base
m	NAV_LAND (21)	0	Final landing

5.5 Inter-Fleet Route Crossing Detection and Repair

Because bases plan sorties independently, route trajectories could intersect. The planner evaluates all inter-base route segments for pairwise geometric intersections. When crossings are detected, a three-tier repair hierarchy executes:

1. *Drop Reassignment:* If intersecting drops are within reach of the neighboring base, they are swapped to eliminate crossing geometry.

2. *Leg Rerouting:* Crossing legs are replanned with a clearance buffer treating the other vehicle's trajectory as an obstacle.

3. *Base Nudging:* If necessary, base coordinates are nudged up to 50 px in 5-px increments to alter radial approach vectors.

If crossings cannot be resolved geometrically, the planner issues a status 2 safety abort, preventing conflicting missions from being flown.

5.6 Geodetic Conversion and Mission Encoding

Route coordinates (x, y) are transformed to geodetic coordinates (φ, λ) relative to image center $\mathbf{c} = ((W'-1)/2, (H'-1)/2)$ and reference (φ_0, λ_0) via local equirectangular projection:

$$E = (x - c_x)\, k, \qquad N = (c_y - y)\, k, \tag{9}$$

$$\varphi = \varphi_0 + \frac{N}{L}, \qquad \lambda = \lambda_0 + \frac{E}{L \cos \varphi_0}, \tag{10}$$

with $L = 111\,320$ m/deg and $k = \rho/s$. Strict coordinate bounds check against polar singularity and meridian wrapping before writing files.

Each vehicle's route is encoded as a standard QGroundControl WPL 110 mission file [38] (Fig. 4, Table 3). At each relief drop, the UAV loiters 5 s at cruise altitude (100 m), descends to delivery altitude (10 m), actuates payload servo channel 9 (2000 μs PWM via `DO_SET_SERVO`), and climbs back to cruise altitude. Reload stops execute `NAV_LAND` and `NAV_TAKEOFF` sequences at base.

5.7 Computational Complexity

With image area $A = W'H'$, morphology and contour extraction require $O(A)$ time. Computing shortest-path fields on the $V_g \approx A/9$ routing grid costs $O(n_{\text{drop}} \cdot V_g)$. Base selection evaluates candidate subsets in $O(K \cdot n_{\text{drop}}^2)$, and sortie generation requires $O(n_{\text{drop}} \log n_{\text{drop}})$. Pairwise crossing checks scale quadratically with total trajectory segments $O(S^2)$. Feasible multi-base fleet plans generate in under 3 minutes on consumer hardware.

6 Verification Methodology

6.1 Independent geodetic oracle

A test-only WGS84 oracle implements Vincenty's direct and inverse solutions without importing production code [44]. It is checked against a published worked geodesic to 1 mm and $0.01''$, an equatorial degree of longitude and the WGS84 meridian quadrant. A pixel's reference position is the point reached from the supplied centre by a geodesic of length $\sqrt{E^2 + N^2}$ at azimuth $\mathrm{atan2}(E, N)$. Model error is bounded using the ellipsoid's radii of curvature. Randomised forward/inverse round trips test valid and rejected inputs.

6.2 Mission validator

An independent specification-based validator reads each exported WPL file and checks item order, command parameters, coordinate ranges, drop actions and pixel-to-coordinate correspondence. It does not import production mission code. Thirty-six corrupted-mission cases exercise the validator itself. Passing it establishes format and declared-model consistency, not autopilot acceptance.

6.3 Metamorphic, mutation and fuzz tests

The same synthetic map is run through the entry point at nine display widths to check ground-position consistency. Mutation testing injects 33 plausible defects in a copy of the code, including axis and scale errors, servo-slot mistakes, a disabled crossing gate and broken reload bookkeeping. Seeded random grids, pathological end-to-end inputs and 500 multi-base scenarios check regression invariants. The reported outcomes are drawn from the recorded test runs.

6.4 SITL and ground-station protocol

An official ArduCopter 4.7.1 SITL binary receives the exported WPL mission through the MAVLink mission-upload protocol. A short mission made by the same production exporter is then flown in AUTO mode to observe takeoff, drop-altitude arrival, servo 9 output, RTL, an intermediate LAND/TAKEOFF pair and final landing. Mission acknowledgements, item progress, relative altitude, servo PWM, position, arming state and battery telemetry are recorded. Separately, MAVProxy loads, uploads, queries and downloads the full WPL through its ground-station command interface; the downloaded item count and command sequence are compared to the export. The full map mission is uploaded but not flown; a separate QGroundControl or Mission Planner graphical-interface test was not performed. Section 8.12 reports the observed outcomes.

7 Experimental Setup

7.1 Data

No public benchmark combines flood masks, delivery truth and UAV trajectories for the complete task of Section 3. Three supplied annotated graphics are ungeoreferenced demonstration inputs (Table 4); two show red flood dots over web maps and Assam is a thematic hazard map. Their PNG resolution metadata do not give ground scale or location. For independent perception and coordinate checks, we use the georeferenced Sen1Floods11 hand-labelled split [4]. One label is rendered as a red overlay for a separate seven-drop pipeline run; because the label constructs that input, it is not an independent HSV accuracy test. Copernicus EMS delineations provide two observed flood transitions [47]. Synthetic maps check software geometry (Section 8.9).

7.2 Setup and Protocol

The original demonstration-map and fleet sweep runs used an Intel Core i5-12500H laptop with 31.7 GiB RAM under Windows 11, Python 3.11.9, NumPy 2.4.6 and OpenCV 5.0.0. New baseline, robustness and sensitivity runs used an AMD Ryzen 7 7840HS laptop with 14 GiB RAM under Linux, Python 3.14.7, NumPy 2.5.2 and OpenCV 5.0.0. Both used the evaluated pipeline defaults (Table 5). Only three substitutions are made, all outside the pipeline code. Colour sampling is performed by a deterministic *simulated operator* that clicks the centroids of the $K = 8$ largest connected components of strongly saturated red pixels ($H \leq 5$ or $H \geq 175$; $S, V > 180$). Display windows are suppressed and their contents captured. The mission is written to the experiment folder instead of the default output path. The three demonstration maps use the default geographic configuration, $(\varphi_0, \lambda_0) = (25.3176^\circ, 82.9739^\circ)$ and $\rho = 2.0$ m/px. All experiments evaluate the unified multi-base fleet framework across base caps $K \in \{1, 2, 3, 4\}$, where $K = 1$ serves as the single-base baseline under identical battery and payload models. All scripts, raw results and figure generators accompany the paper (Data Availability). The Linux Kanpur $K = 4$ moderate rerun serves 35/41 drops, while the earlier Windows fleet sweep recorded 34/41; these are reported as separate platform runs, without attributing the difference to a specific cause.

Table 4: End-to-end inputs. Only the Sen1Floods11 diagnostic has a GeoTIFF reference; its red annotation is rendered from the hand label.

Map	Size (px)	Display	Content and printed source
Varanasi	972 × 970	750 × 748	red flood dots on web basemap; "ESA data . . . processed by India Today"
Kanpur	976 × 966	750 × 742	red flood dots on web basemap; "Sentinel-2 SAR data . . . ESA" (sic)
Assam	1048 × 739	750 × 529	thematic hazard map (5 classes, rivers, roads); no source printed
Ghana S1/S2	512 × 512	512 × 512	Sen1Floods11 label rendered red; GeoTIFF CRS/affine

Table 5: Pipeline configuration used in all experiments (repository defaults). The fleet evaluation additionally uses a moderate weather preset (3 mm, 12 km/h, 270°).

Parameter	Value
Display width D; min. area $a_{\min}$	750 px; 200 px^2
HSV patch; δ_H, δ_{SV}; element B	11 × 11; 20, 30; 5 × 5
Weather (P, V, θ)	10 mm, 30 km/h, 270°
Horizon T; spread coefficients	2 h; 5 mm/px, 10 km/h/px
Grid factor; snap radius	3; 4 cells
Reference (φ_0, λ_0)	25.3176°, 82.9739°
Scale ρ	2.0 m/px
Cruise / drop altitude; loiter	100 / 10 m; 5 s
Servo channel; PWM	9; 2000 μs
Fleet and battery parameters (Section 5)	
Base cap K; separation; clearance	6; 150 px; 5 px
Speed v; endurance E; c_m; c_w	5 m/s; 7 min; 0.08/kg; 0.025 s/m
Payload per drop; capacity; reserve r	0.25 kg; 8 kg; 20%
Contour tiers $(a_{\mathrm{mod}}, a_{\mathrm{large}}, a_{\mathrm{huge}})$	2000, 8000, 25 000 px^2
Drop sep. s_{drop}; cap k_{drop}; step	50 px; 4; 3000 px^2

7.3 Baselines

Perception baselines are Otsu, k-means, NDWI and a small U-Net on labelled S1/S2 imagery. Flood spread is compared with persistence; ordering with exact and heuristic tours; D* Lite with A*; and fleet allocation with balancing and OR-Tools. Inputs and constraints are matched where feasible.

7.4 Metrics

We report flood-mask and obstacle fractions, regions and deliveries, no-path legs, route and mission size, and runtime. The labelled benchmark reports pixel-pooled IoU, F1, precision and recall plus mean chip IoU. The forecast study uses next-frame IoU against persistence. The route ablation uses exposure (route fraction over an obstacle set, sampled every 0.5 px). Fleet metrics include coverage, optimistic longest-route cruise

Table 6: Held-out Sen1Floods11 hand-labelled chips. Metrics pool labelled pixels; mean-chip IoU exposes site variation. Lab k-means has 87 valid chips, all others 88.

Method	IoU	F1	Precision	Recall	Mean IoU
S1 VH Otsu	.225	.368	.246	.732	.219
S1 VH k-means	.226	.369	.246	.734	.221
S1 two-band U-Net	.606	.755	.766	.744	.300
S2 RGB Otsu	.086	.159	.095	.485	.084
S2 Lab k-means	.096	.175	.103	.586	.095
S2 NDWI> 0	.772	.871	.923	.826	.474

time, route-load Gini coefficient and maximum estimated sortie battery fraction. Geographic error is geodesic distance to the supplied affine/CRS coordinates.

8 Results and Discussion

8.1 Georeferenced Flood Benchmark

We used the official Sen1Floods11 hand-labelled train/validation/test lists (251/86/90 chips before exclusions) [4]. Two test chips had no finite Sentinel-1 pixels at valid labels, leaving 88. Fixed baselines include Otsu and k-means on Sentinel-1 VH; a dark-cluster Lab k-means and grayscale Otsu on Sentinel-2 RGB; and NDWI> 0 on Sentinel-2 green/NIR [13]. Lab clustering was undefined on one constant-colour chip. A three-level two-band S1 U-Net was trained for 12 epochs at 128×128 px, selected by validation IoU (epoch 11), and evaluated at the original resolution. Cluster choices use brightness, not held-out labels.

The S2 index performs best, while unsupervised brightness clustering fares poorly. Sensor differences prevent treating the cross-sensor gaps as controlled algorithm comparisons. The production HSV operator detects *red annotations*; its red-click sampler found no applicable flood marks in raw S2 RGB, so no raw-image IoU is assigned. Supplementary Fig. S1 shows two held-out examples, including one with weak U-Net transfer.

The GeoTIFF affine and CRS map 81 locations per test chip to WGS84. With the scalar converter anchored at each chip centre and set to mean local pixel size, the median chip-maximum error is 158.5 m, the 95th percentile 406.5 m and the global maximum 419.5 m. An optional GeoTIFF input path now transforms mission pixels through the full affine and CRS. On a 512-px Ghana chip, a red rendering of the hand label produced seven drops and a 197-item mission; its navigation coordinates matched the independent affine calculation to displayed precision, versus 24.5 m maximum scalar-route error. The rendering occupies a different input domain from raw satellite imagery, and its 50.6% HSV mask is not an independent segmentation score. Supplementary Fig. S3 overlays the route on the co-registered Sentinel-2 image; this is a geographic product overlay, not surveyed flight accuracy.

8.2 End-to-End Results

Table 7 and Fig. 5 summarise the end-to-end runs. All four runs complete and produce missions that pass the validator with no errors or warnings. Path planning accounts for more than 99% of runtime; every other stage takes less than 0.1 s. The fourth run is a label-rendered georeferenced diagnostic rather than an independent flood-detection evaluation. Three findings qualify these outcomes.

1. *No-path legs.* Half of the Varanasi legs (8 of 16) and 6 of 33 Kanpur legs have no path on the planning grid. The mission therefore flies straight segments across obstacles on these legs (red segments in Fig. 5). The independent A* baseline (Section 8.6) confirms that these endpoints are genuinely disconnected on the grid: after snapping, they lie in free pockets enclosed by the inflated flood mask.

2. *Segmentation failure on the thematic map.* For Assam, the sampled patches include white background, ℓ_S falls to 0 and achromatic pixels pass the red hue test (Section 4-B). Of the image, 83.1% is labelled flood, the whole map becomes a single region, and both endpoints of the only (zero-length) leg remain inside obstacles. The resulting 10-item mission is structurally valid but meaningless.

3. *Geography.* All missions lie within about 2 km of the default reference, irrespective of the map content (Section 8.10).

Table 7: End-to-end results (simulated operator). The Ghana column uses a label-rendered annotation, calm weather and the optional GeoTIFF converter; the other columns use the default configuration.

	Varanasi	Kanpur	Assam	Ghana
Flood-mask fraction M (%)	8.10	15.08	83.10	50.59
Obstacle fraction O (%)	15.81	29.06	92.24	50.59
Regions = deliveries	16	33	1	7
Planned legs / no-path legs	16 / 8	33 / 6	0 / 0	7 / 0
Endpoints snapped / unsnappable	30 / 0	68 / 0	0 / 4	8 / 0
Route points	297	1010	3	166
Mission items	364	1145	10	197
Route length at configured scale (km)	5.66	18.78	0.00	9.34
Validator errors / warnings	0 / 0	0 / 0	0 / 0	0 / 0
Total runtime (s)	13.4	35.7	0.03	2.22
Share of runtime in planning	99.4%	99.6%	–	99.5%

Table 8: Ablation of predicted-flood obstacles. Exposure is the share of route length over the given area.

Map	Obstacles	No-path	Items	Time (s)	Exposure pred.-only	Exposure current
Varanasi	M	0	424	4.7	33.8%	3.4%
	$M \cup \hat{M}$	8	364	13.4	17.9%	4.8%
Kanpur	M	0	733	16.5	47.9%	5.5%
	$M \cup \hat{M}$	6	1145	36.4	11.6%	4.3%

An optional excessive-mask guard at 80% stops Assam before export (exit status 2), while Varanasi and Kanpur still produce missions. The threshold is a safety diagnostic, not a calibrated classifier; Supplementary Fig. S6 shows the failed mask and guard outcome.

8.3 Ablation: Predicted-Flood Obstacles

Table 8 compares planning on $O = M \cup \hat{M}$ with planning on M alone. Predicted-only pixels cover 7.7% (Varanasi) and 14.0% (Kanpur) of the image. Without predicted obstacles, 33.8% and 47.9% of the route length lies over predicted-only area. With them, exposure falls to 17.9% and 11.6%. Exposure does not reach zero because the no-path legs fly straight. The same inflation, however, causes *all* no-path legs: there are none with M alone, against 8 and 6 with O. It also increases planning time by a factor of 2.2–2.9, and for Varanasi it slightly increases exposure to the current flood (3.4% to 4.8%). The prediction stage is therefore not uniformly beneficial: in its current form, the benefit it is designed for comes with a loss of planner completeness.

8.4 Observed Flood Progression and Operator Robustness

Two successive delineation pairs for one Copernicus EMSR504 Australian flood area were rasterised on a common 30 m grid and compared with persistence and the project's weather-driven dilation [47]. Retrospective hourly precipitation and wind at the first acquisition came from the Open-Meteo historical reanalysis [48]. The 19–21 March 2021 pair has 2-day observed-mask IoU 0.803 for both persistence and the proxy; the 21–22 March pair has IoU 0.946 for both. At 0 and 0.4 mm/h precipitation respectively, the proxy's integer spread threshold generated no change, while mapped flood cells changed. Thus these two observed comparisons provide no evidence of forecast improvement. The reanalysis is retrospective

and mapping products have observation uncertainty. Supplementary Fig. S2 shows the two observed transitions, the unchanged proxy and their disagreements.

To sample the interactive stage, we ran ten deterministic simulated operators per map under calm, moderate and heavy weather, varying clicks from four to eight and perturbing click locations by at most 3 px (90 full runs). The Varanasi and Kanpur mask fractions were 8.55±0.63% and 14.56±1.19% (mean±sample SD). Moderate weather produced no no-path legs; heavy weather produced 8.2±0.6 and 6.0±0.5 respectively. Kanpur route length was 3495±239, 3637±266 and 6580±504 display px under calm, moderate and heavy conditions. The Assam mask remained 83.09% and yielded a structurally valid but unusable mission in every run. These are simulated click perturbations, not measurements of human variability. Supplementary Fig. S5 shows route-length distributions by weather.

8.5 Sensitivity

A deterministic sweep varies the colour margins and number of operator clicks. The hue margin $\delta_H \in \{10, 20, 30\}$ has *no* effect on any map, because the red-wrap rule replaces the hue interval. The saturation/value margin $\delta_{SV} \in \{15, 30, 45\}$ has a moderate effect on the two dot maps, while the Assam result is insensitive to every setting: its contamination is structural rather than a tuning artefact. Using 16 instead of 8 clicks raises the Varanasi flood fraction from 8.1% to 9.7% and the region count from 16 to 23, so the interactive step is a material source of variability. The ground scale ρ affects only the geodetic conversion: the Varanasi route length scales linearly at 2.83 km per m/px, so the 5.66 km mission at $\rho = 2$ m/px would be 263 km at the landmark-estimated scale of the map (Section 8.10).

8.6 Route Ordering and Planner Baseline

Two evaluation-only baselines were tested; neither changes A.D.A.P.T.. Against exact optima from Held–Karp dynamic programming [29] on uniformly random instances, the nearest-neighbour order is on average 4.6% (5 points) to 12.0% (12 points) longer, with a worst case of 44.5%. On the 16 Varanasi drop points it is 3.2% longer than optimal. For the 33 Kanpur points only a minimum-spanning-tree lower bound is available, which limits the gap to at most 37.8%. On the same actual points, evaluation-only 2-opt shortens the 33-point Kanpur closed Euclidean tour from 2934 to 2458 display px (16.2%); Or-opt and an OR-Tools tour give the same length, without proving optimality. On Varanasi, Or-opt and OR-Tools reach the Held–Karp optimum of 1669 px versus nearest neighbour at 1722 px. These ordering lengths exclude obstacle detours.

A plain grid A* [24] agrees with D* Lite on reachability for all 49 non-trivial legs and on path cost to within 2×10^{-13}. The evaluated D* Lite runs are 61–112 times slower (12.67 versus 0.21 s on Varanasi; 30.83 versus 0.28 s on Kanpur) because each leg rebuilds the search instance instead of exploiting incremental reuse.

8.7 Fleet Scaling and Base-Cap Performance (From Baseline $K = 1$ to $K = 4$)

Protocol. The multi-base fleet planner (Section 5) was evaluated on the benchmark maps under dummy weather and moderate weather presets (3 mm, 12 km/h, 270°) across base caps $K \in \{1, 2, 3, 4\}$ (24 runs). For a given map and weather, all caps share the drop points, dynamic obstacles and vehicle parameters, so K is the independent variable. $K = 1$ serves as the baseline single-base case where one vehicle flies all sorties from a single launch base. All metrics are computed directly from each run's outputs, and every exported mission was validated against its route file. The *estimated completion time* is the routed length of the longest mission divided by cruise speed $v = 5$ m/s, representing an optimistic lower bound excluding ground turnaround. Table 9 gives the results.

Feasibility and safety checks. Of the 24 runs, 20 produced missions, 43 in total. In these runs:

- all 43 missions pass the validator with no errors;
- no two routes of a run cross, so no repair was needed;
- no leg was flown as a straight fallback, and no route pixel lies in the detected flood;
- every visited drop lies in its own base's territory;
- the largest routed sortie uses 79.6% of the battery, against the 80% limit.

The four Assam runs under moderate weather end with status 2, because no base site reaches any drop. Under dummy weather, Assam has one base and visits 1 of 4 drops for every K. Both outcomes follow from the segmentation failure of Section 8.2.

Coverage. A base can only serve drops within one single-drop sortie (8). That is about 832 m of routed distance in calm air, 768 m under the moderate wind and 690 m under the dummy wind, or roughly 300 and 270 display pixels at the Varanasi scale. Adding bases therefore mainly adds coverage. From $K = 1$ to $K = 4$, the visited drops rise as follows:

- under dummy weather, from 6 to 12 of 22 (Varanasi) and from 10 to 23 of 41 (Kanpur);
- under moderate weather, from 10 to 18 and from 17 to 34.

Every unvisited drop is reported. In all but two runs, the reason is that no admissible base reaches it within the cap; in two Kanpur runs, the sortie planner rejected one assigned drop. The stronger dummy wind shortens the reach and lowers coverage at every K. Because the visited sets differ, total route length does not compare efficiency like for like. It grows with K because more drops are served and each base flies its own return legs.

Clustering. Across the 12 multi-drone runs, 190 of 206 served regions were assigned to one drone; 16 large regions were shared by two.

Workload. The nearest-feasible assignment does not balance work. On Kanpur under moderate weather, the drops per UAV are as follows:

- $K = 1$: 17;
- $K = 2$: 17 and 8;
- $K = 3$: 17, 7 and 6;
- $K = 4$: 15, 7, 6 and 6.

The longest mission therefore stays at 11.0 km, and the completion estimate at 37 min, up to $K = 3$; the estimate falls to 28 min only at $K = 4$. On Varanasi under moderate weather, the longest mission falls from 10.3 to 6.4 km and the completion estimate from 34 to 21 min, using three of the four permitted bases.

Choosing an approach. The planner uses one base whenever it reaches every reachable drop; otherwise adding bases raises coverage. The single-UAV approach visits all 16 Varanasi and 33 Kanpur regions but has no endurance check. Its routes of 5.66 and 18.78 km exceed the fleet model's approximate 2.1 km per-battery cruise allowance. Thus $K = 1$ in the fleet mode is the battery-constrained one-drone comparison.

Scale dependence. These coverage figures hold only for the configured scale. The $K = 4$, moderate-weather runs were repeated at the landmark-estimated scales of Section 8.10 (93 and 130 m/px). At those scales a single-drop sortie spans only 6.4 and 4.5 display pixels, every drop is unreachable, and both runs end with status 2.

Runtime. A feasible run took between about 10 s and 3 min on the evaluation laptop. These are single measurements, taken with several runs executing in parallel.

Integration equivalence. Three identical-input reference comparisons (both maps in calm conditions and Varanasi with

Table 9: Multi-base autonomous fleet planning across base caps $K \in \{1, 2, 3, 4\}$ on benchmark flood relief missions (simulated operator, `METERS_PER_PIXEL`= 2 m per original pixel). $K = 1$ serves as the baseline single-base case. Visited: drops delivered of dry drop points generated. km: routed length of all missions; max/UAV: longest single mission. Time: estimated completion (minutes) from route length at 5 m/s, all UAVs launched together, excluding take-off, landing, drop manoeuvres and reload turnaround. Every exported mission passed the independent validator; no two routes cross.

		Dummy weather (10 mm/h, 30 km/h)						Moderate weather (3 mm/h, 12 km/h)					
Map	K	UAVs	Visited	Sorties	km	max/UAV	Time	UAVs	Visited	Sorties	km	max/UAV	Time
Varanasi	1	1	6/22	3	3.0	3.0	10	1	10/22	8	10.3	10.3	34
	2	2	9/22	5	5.3	3.0	10	2	16/22	10	13.0	7.6	25
	3	3	11/22	6	6.3	2.3	8	3	18/22	10	12.9	6.4	21
	4	4	12/22	6	5.7	2.1	7	3	18/22	10	12.9	6.4	21
Kanpur	1	1	10/41	3	3.1	3.1	10	1	17/41	9	11.0	11.0	37
	2	2	16/41	5	5.4	3.1	10	2	25/41	14	17.2	11.0	37
	3	3	21/41	8	8.7	3.2	11	3	30/41	16	19.4	11.0	37
	4	4	23/41	9	9.4	3.2	11	4	34/41	16	19.6	8.5	28
Assam	1	1	1/4	1	0.5	0.5	2	infeasible: no base reaches any drop (exit 2)					
	2	1	1/4	1	0.5	0.5	2	infeasible: no base reaches any drop (exit 2)					
	3	1	1/4	1	0.5	0.5	2	infeasible: no base reaches any drop (exit 2)					
	4	1	1/4	1	0.5	0.5	2	infeasible: no base reaches any drop (exit 2)					

Table 10: Linux $K = 4$ allocation comparison. Coverage is constant within each row group; time is optimistic longest-route cruise time.

Map, weather	Allocation	Drops	Time (min)	Gini
Varanasi, dummy	nearest	12/22	7.05	.239
	balanced	12/22	6.41	.082
Kanpur, moderate	nearest	35/41	28.23	.230
	balanced	35/41	21.77	.098
	OR-Tools	35/41	28.57	.198
Varanasi, moderate	nearest	18/22	21.38	.271
	OR-Tools	18/22	26.09	.344

moderate weather) reproduced base positions, assignments, routes and drop/reload indices exactly. The reference rejects the Assam mask before planning.

8.8 Allocation and Model Sensitivity

We compared nearest-feasible allocation with a routed round-trip workload proxy and an OR-Tools multi-depot allocation baseline. Both are evaluation-only substitutions at the assignment stage; production routing, sortie construction and WPL validation remain unchanged. OR-Tools does not enforce the full downstream weather, battery or crossing model, so it is not a certified optimum for this problem. All compared missions passed the validator with no route crossings.

Balancing improves the longest-route estimate by 9.1% and 22.9% in its two responsive cases, while increasing total flown route length from 5.70 to 6.83 km and from 18.75 to 20.85 km. The other two map/weather cases are unchanged. OR-Tools does not improve the objective after the production feasibility checks and routing. Supplementary Fig. S4 plots per-vehicle workloads and the associated makespan/coverage comparison.

At $K = 4$ with moderate weather, varying the assumed scale across 2, 3, 5, 10, 20 and 50 m/px serves 18, 14, 11, 6, 3, 0 of 22 Varanasi drops, and 35, 27, 19, 8, 4, 0 of 41 Kanpur drops; landmark-estimated 93 and 130 m/px also serve none. Reducing model endurance by 20% cuts coverage to 16 and 30, while increasing it by 20% leaves coverage unchanged but can cause selection of fewer bases and a longer estimated makespan. Increasing the wind penalty by 20% reduces Kanpur coverage from 35 to 32. A 20% one-at-a-time change in per-drop payload mass changes neither map's coverage. These are model perturbations without flight-log calibration. Single wall-time measurements are recorded with each scale run; apparent runtime reductions at coarse scales reflect early rejection of unreachable drops, not a general complexity result. Supplementary Fig. S5 collects the battery, scale and runtime sweeps.

8.9 Geospatial and Mission Verification

Table 11 summarises the verification results. Every result in this subsection concerns *software correctness*, that is, whether the code computes what its model specifies. None of it shows that the generated coordinates match real places.

- *Synthetic ground truth.* On a 1000×1000 px synthetic image at 10 m/px, the implementation matches an independent re-derivation of (9)–(10) to within 1.4×10^{-14} degrees at nine control points. Against the WGS84 geodesic ground truth, the largest error is 25.9 m at the image corners, dominated by the 0.49% north–south scale error of L at 25°, and every point is within the derived bound.

- *Round trips and edge cases.* Of 100 000 randomised round trips through a test-only inverse, covering mid-latitudes, near-polar and near-antimeridian references, image sizes and scales from 0.05 to 633 m/px, 98 789 are representable and agree within 6.8×10^{-8} px; the remaining 1 211 are correctly rejected.

- *Mission correspondence.* Every row of the three missions (1 519 rows in total) matches an independent conversion of its source pixel within 3.6×10^{-15} degrees. Traced back to the original image, the deviation is at most 0.57 m, entirely north–south and explained by the rounding of H'.

- *Resize consistency.* A synthetic three-region map run at nine display widths (300 to 2400 px, odd and even) yields drops at the same physical corners. They deviate from the model by at most one display pixel, and by 7×10^{-10} m when no resizing occurs.

- *Mutation testing.* 31 of 33 injected defects were detected in the earlier 231-test mutation campaign. Of the 24 defects in the geospatial and mission layers, 22 were detected; all 9 defects injected into the multi-base integration were detected, for example an uncompensated display scale, a disabled crossing gate, a missing reload stop or a stale reach set after nudging. The two surviving mutants are behaviour-equivalent: one removes a redundant finiteness guard, the other repeats a conversion.

- *Adversarial testing.* Property-based tests of D* Lite and crossing checks, about 110 pathological end-to-end runs and 500 seeded fleet scenarios exposed ten defects, all corrected before the reported evaluation. None changes the missions reported above.

  None of the corrections changes a mission generated in the experiments above.

During verification, the procedure exposed and led to the correction of defects in the original implementation: scale was not compensated after resizing, servo parameters were in the wrong slots, the region chosen as HOME received no delivery, the reference pixel was off by half a pixel, and invalid or polar inputs silently produced invalid coordinates. All results reported here use the corrected code.

Table 11: Software verification of the geospatial and mission layers, and overall validation status.

Check	Result
Software correctness (performed)	
Automated tests (incl. adversarial)	238 + 520 subtests pass
Oracle vs. published geodesic	$\leq$1 mm, $\leq 0.01''$
Implementation vs. model (synthetic)	$\leq 1.4 \times 10^{-14}$ deg
Model vs. WGS84, $\leq$7.1 km offsets, 25°	$\leq$25.9 m
Round trips (98 789 compared)	max 6.8×10^{-8} px
Mission rows vs. source pixels (1 519)	$\leq 3.6 \times 10^{-15}$ deg
Resize, 9 display widths	$\leq$1 display px
Mutants detected	31/33 (2 equivalent)
Validator attack cases	36 pass
Single-UAV missions (4 maps)	0 errors, 0 warnings
Multi-base missions (20 runs, 43 files)	0 errors, no crossings
Multi-base plans vs. reference (3 cases)	identical
Fuzzed end-to-end scenarios (500)	0 invariant violations
Validation beyond software correctness	
Geographic validity of supplied PNG maps	not established
GeoTIFF diagnostic mission (197 items)	0 errors, 0 warnings
MAVProxy upload / download	364 / 364 items accepted
QGroundControl / Mission Planner GUI	not performed
ArduCopter SITL upload / short mission	accepted / flown
Hardware-in-the-loop / physical flight	not performed

8.10 Geographic Validity of the Supplied Maps

To test whether the configured geography could describe the maps, a similarity transform was fitted between visible place-name labels and approximate public coordinates of those towns. This was done for 5, 10 and 12 labels on the Varanasi, Kanpur and Assam maps; the label positions were read off by eye. The estimate is a plausibility check, not a georeference. Residuals are 15–26 px and the rotations are $-0.5°$, $-1.0°$ and $-4.8°$. The fitted scales are about 93, 130 and 633 m/px, which is 46, 65 and 317 times the configured 2.0 m/px. The default reference lies 23 km, 319 km and about 1 000 km from the estimated image centres. At those extents, the flat-earth model itself would err by up to 345 m, 563 m and 7.7 km (from the WGS84 oracle), and the maps' web-Mercator basemaps violate the uniform-scale assumption [43]. The generated missions are therefore *not* geographically meaningful for these maps. The cause lies in the inputs and configuration, not in the conversion code.

8.11 Mission Semantics

The command parameters were audited against the MAVLink definitions and the ArduPilot and PX4 documentation [39–41, 49]; the findings are given in Section 4-K and Section 9. The lower block of Table 11 separates measured acceptance from remaining operational checks. The multi-base reload LAND/TAKEOFF semantics are evaluated in the short ArduCopter SITL run of Section 8.12: LAND disarms and resets the mission index, so automatic continuation is unsupported by this configuration [50]. The optional GeoTIFF route coordinates are independently recomputed from the reference affine transform and CRS; the Ghana mission has zero observed navigation-coordinate discrepancy at WPL precision.

8.12 Simulation and Mission Execution

ArduCopter 4.7.1 SITL accepted all 364 items of the exported Varanasi WPL via the MAVLink mission protocol (364 item requests; acceptance acknowledgement), and accepted a 17-item two-drop/reload WPL from the same exporter [40, 49]. MAVProxy independently loaded the 364-item WPL, received an accepted MISSION_ACK, reported 364/364 items present, and downloaded 364 items with the same command sequence; its six-decimal coordinate export differed by at most 5.98×10^{-7} degrees from the input. Supplementary Fig. S7 is an excerpt from that actual ground-station transcript. The short WPL was executed in AUTO at fivefold simulated speed. Telemetry recorded arrival at 9.98 m relative altitude and servo 9 changing from 1000 to 2000 μs at the first drop; the intermediate LAND disarmed the vehicle. ArduCopter reset its mission index to TAKEOFF item 1 rather than continuing automatically at item 8. A guided rearm alone repeated the first sortie. Explicitly selecting item 8 after rearm reached the second drop; RTL landed the vehicle and the final LAND item became current, without a recorded completion event. The second DO_SET_SERVO command was reached, but servo 9 remained at 2000 μs because the exporter emitted no reset between drops. Thus a second release is not established. The SITL battery-voltage reading remained at 12.6 V and cannot calibrate the energy model. These outcomes come from telemetry, not WPL structure. The full 364-item route was uploaded but not flown. This is one simulated vehicle, with neither a QGroundControl/Mission Planner graphical-interface test nor a physical flight. SITL does not validate the map geography, landing-site safety or payload release mechanism.

9 Limitations and Threats to Validity

Input validity. The three supplied PNG maps have neither georeferences nor labelled flood truth. Their configured centre and scale do not match the depicted geography (Section 8.10), so exported coordinates cannot be used for field missions. The optional GeoTIFF converter corrects coordinates only when its raster is co-registered to the annotation; it does not create ground truth or verify registration. The Ghana diagnostic uses a hand label to construct the red input and therefore cannot validate HSV detection. The supplied maps' HSV masks extract annotation colours, not water from raw imagery. The separate Sen1Floods11 benchmark tests S1/S2 flood segmentation, but its sensor methods cannot be treated as direct replacements for the annotation detector without a changed input pipeline. The Assam thematic map is an observed failure: 83.1% of pixels are marked flood; an optional 80% guard rejects it, but that threshold is uncalibrated. The ten-seed click study represents simulated operators, not measured people.

Forecast validity. The spread rule is a terrain-free dilation proxy. It ignores drainage and elevation, and assumes a weather interval that may differ from the source API's accumulation interval [51]. In the two EMSR504 observed transitions, it equals persistence rather than improving upon it; both transitions belong to one event. Retrospective reanalysis weather is not an operational forecast, and delineation maps have uncertainty. Lower route exposure to the proxy's own predicted area is not flood forecast skill.

Route and mission validity. Single-UAV planning samples one cell per 3×3 block, can place drops in blocked cells, and flies straight when a leg is disconnected. The 14 observed no-path legs remain unsafe, and the mode has no endurance check. Multi-base drops are dry only under the computed mask, with no landing-site or crowd-safety assessment. The battery coefficients and assumed wind penalty lack flight-log calibration. Balanced assignment improves the longest-route proxy in two cases but increases total route length. A route crossing check provides no separation in time or altitude.

Study design. End-to-end results use three convenience graphics and one label-rendered georeferenced diagnostic, while labelled segmentation uses 88 held-out Sen1Floods11 chips. Geospatial chip errors measure a scalar approximation of supplied GeoTIFF transforms, not field position accuracy. The two flood transition pairs come from a single emergency activation. Platform-specific fleet reruns differ by one Kanpur drop (34 versus 35 of 41); the cause is unresolved. Runtimes are single laptop measurements and are not algorithmic scaling estimates. The related-work search was targeted rather than systematic.

Operational validation. The full map WPL was accepted in ArduCopter SITL and uploaded/downloaded through MAVProxy, but not flown; a short WPL produced one servo transition, showed intermediate LAND resets the mission index, and reached the second drop only after an explicit restart. With no servo reset, a second release was not demonstrated. No QGroundControl or Mission Planner graphical-interface test, PX4 test, hardware-in-the-loop test or physical flight is avail-

able. Before operational use, the system needs georeferenced input, safe disconnected-leg handling, landing-site assessment, a calibrated vehicle and release mechanism, vehicle failsafes and applicable aviation approval. Multiple vehicles also need temporal or vertical deconfliction.

10 Conclusion

A.D.A.P.T. integrates annotated-map hazard extraction, a weather-driven obstacle proxy, dry relief drop allocation, multi-base fleet routing, pixel-to-geodetic conversion and WPL mission export. The multi-base autonomous fleet planner generates structurally valid, collision-free missions on the demonstration maps across single-base ($K = 1$) and multi-base fleet configurations. Independent geodesic and WPL checks support consistency with the implemented model; an optional GeoTIFF converter provides a raster-derived geographic path, and a mask-fraction guard rejects the observed Assam failure. ArduCopter SITL and MAVProxy accept a full exported mission, while a shorter simulated flight records the first drop action.

The added experiments delimit that result. On 88 labelled Sen1Floods11 test chips, S1 Otsu, a small S1 U-Net and S2 NDWI achieve micro IoU 0.225, 0.606 and 0.772, respectively; the annotation HSV detector is not a raw-image baseline. The flood spread proxy does not beat persistence on two observed EMS transitions. In SITL, intermediate landing resets the mission index and requires an explicit restart; the second servo command does not create a new PWM transition. A balanced assignment lowers the fleet's optimistic longest-route time by up to 22.9% at unchanged coverage, whereas OR-Tools assignment alone does not improve it. Reachability collapses as the assumed metres per pixel increases, and a 20% endurance reduction removes reachable drops. The supplied maps lack a usable georeference and the vehicle model is uncalibrated. The paper therefore provides a reproducible software and simulation baseline; field missions require corrected inputs, a safe route policy and physical validation.

11 Future Work

The immediate engineering priorities are to reject disconnected route legs, reset the servo between deliveries, and verify an explicit reload sequence in ArduPilot and PX4 SITL. A flight-log-calibrated battery model and site-safety assessment are needed before field trials. The optional GeoTIFF conversion should be extended to verified image registration and ground-control points, while direct annotation accuracy needs truth labels on the three demonstration maps. Flood progression should be tested against more independent observed events and terrain-aware models. Finally, allocation should optimise routed fleet completion time while enforcing endurance, payload, and temporal or vertical vehicle separation.

Data and Code Availability

The implementation is hosted at `https://github.com/TSaha4/A.D.A.P.T.-Autonomous-Drone-for-Adaptive-Path-Tracking`. Evaluation scripts, recorded results, WPL files, trained weights and a checksummed source-data manifest accompany the manuscript. The three demonstration maps are third-party graphics without flood ground truth or georeferencing; redistribution rights must be confirmed before publication. The Sen1Floods11 labels and imagery and the Copernicus EMS flood delineations are publicly available from their cited sources. SITL telemetry is included with the evaluation results; no physical flight log is available.

Declarations

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Author contributions. To be completed by the authors before submission.

AI assistance. AI tools assisted manuscript drafting, evaluation scripts, verification tests and integration review. The authors remain responsible for checking every claim, result and disclosure before submission.

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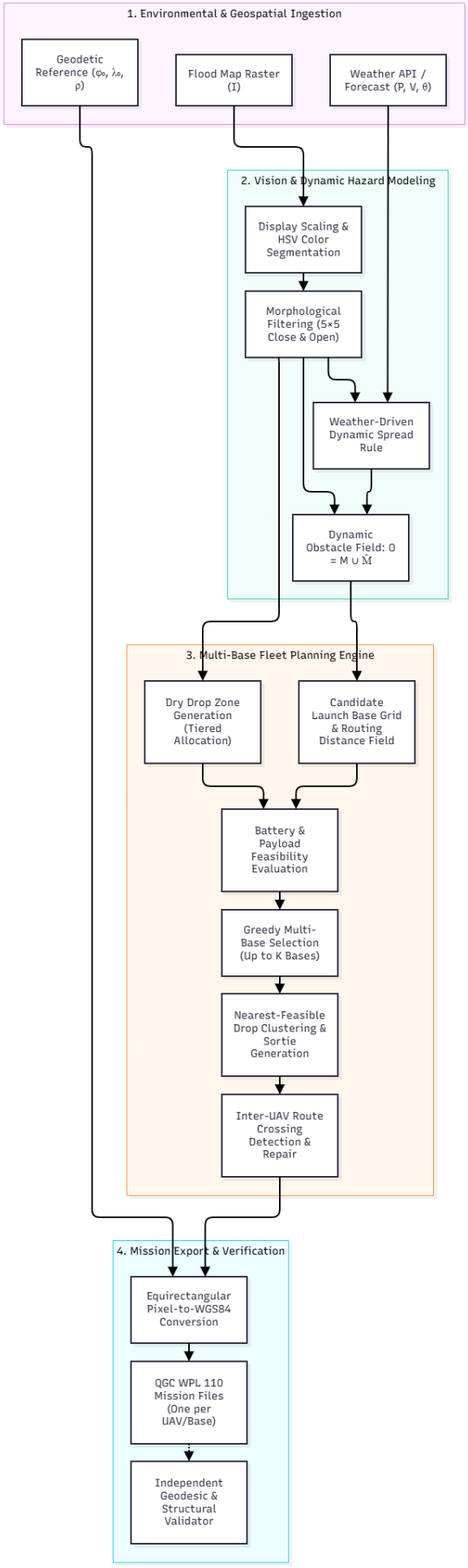


Figure 1: Overall high-level system architecture of the A.D.A.P.T. autonomous multi-base fleet relief pipeline, from environmental ingestion to executable mission export.

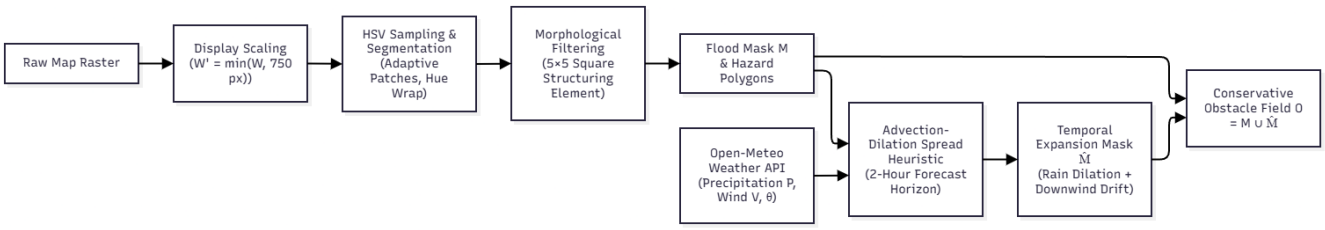
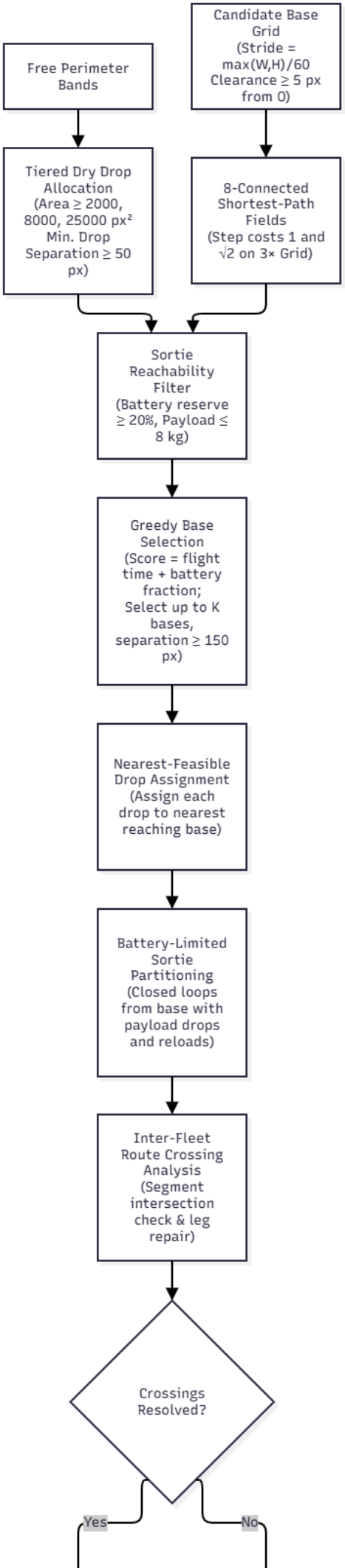


Figure 2: Division 1: Flood perception and dynamic weather spread heuristic, extracting hazard regions and generating the composite obstacle field $O = M \cup \hat{M}$.



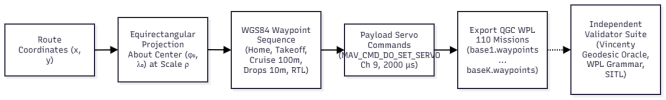


Figure 4: Division 3: Pixel-to-geodetic projection and QGroundControl WPL 110 autopilot mission synthesis.

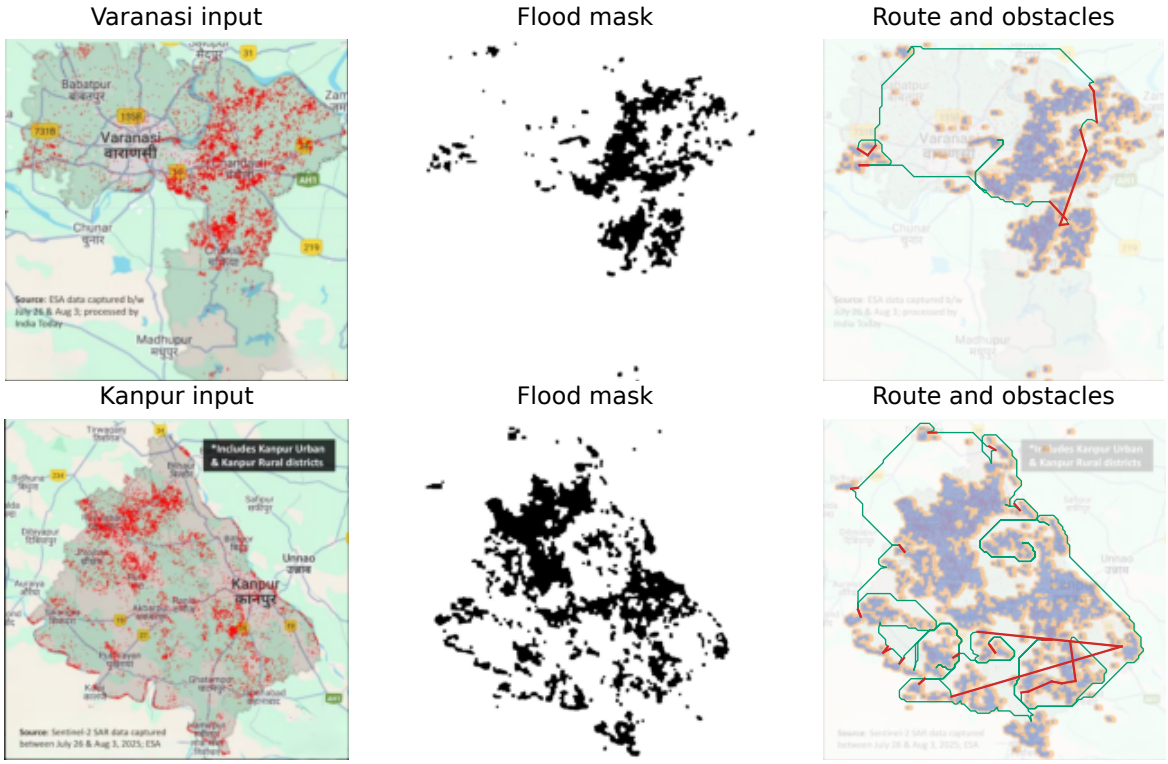


Figure 5: Recorded pipeline outputs for Varanasi and Kanpur: annotated input, extracted flood mask, and planned route over current (blue) and predicted-only (orange) obstacles. Red route segments mark long straight fallbacks; they are not validated flight paths. The third-party input imagery is reproduced for research review only.

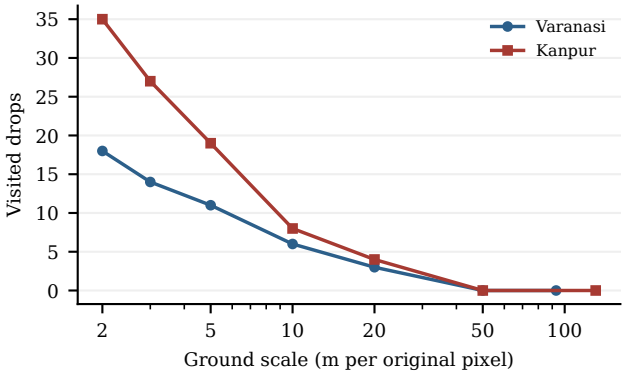


Figure 6: Served drops versus assumed metres per original pixel at $K = 4$ and moderate weather. The supplied PNG maps have no actual georeference.

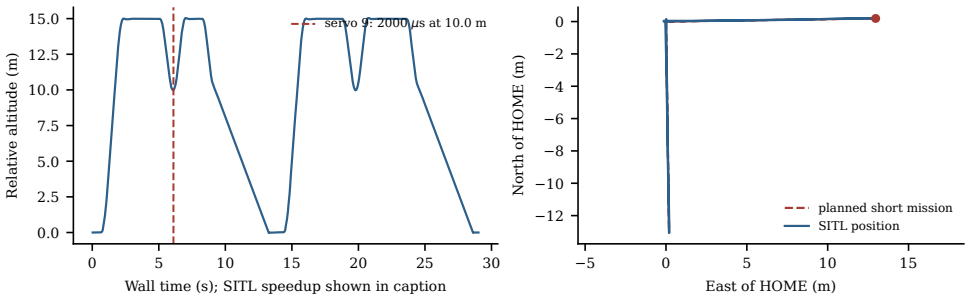


Figure 7: Short exported mission in ArduCopter SITL at 5× speedup. Altitude and servo event come from telemetry; the position panel compares recorded positions with the planned short mission.